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DEMONSTRATE AN UNDERSTANDING OF DIFFERENT COMPUTER NETWORK ARCHITECTURES AND STANDARDS

UNIT STANDARD 114074

NQF LEVEL: 5

CREDITS: 5

NOTIONAL HOURS: 50

LEARNER GUIDE

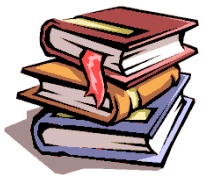
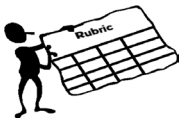
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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 **Books**	**This icon means that other books are available for further information on a particular topic/subject.**
 **References**	**This icon refers to any examples, handouts, checklists, etc…**
 **Important**	**This icon represents important information related to a specific topic or section of the guide.**
 **Activities**	**This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.**
 **Exercises**	**This icon represents any exercise to be completed on a specific topic at home by you or in a group.**
 **Tasks/Projects**	**An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.**

 **Workplace Activities**	**An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace**
 **Tips**	**This icon indicates practical tips you can adopt in the future.**
 **Notes**	**This icon represents important notes you must remember as part of the learning process.**

Learner Guide Introduction

About the Learner Guide…	This Learner Guide provides a comprehensive overview of the **114074 Demonstrate an understanding of different computer network architectures and standards** and forms part of a series of Learner Guides that have been developed The series of Learner Guides are conceptualized in modular's format and developed FOR **NATIONAL CERTIFICATE: INFORMATION TECHNOLOGY: SYSTEMS SUPPORT: SAQA 48573 -LEVEL 5- 147 CREDITS** They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules, and conducted by a qualified facilitator.
Purpose	114074 Demonstrate an understanding of different computer network architectures and standards
Outcomes	• 114074 Demonstrate an understanding of different computer network architectures and standards
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning programme, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency
Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate
Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.

Learning Unit 1

Demonstrate an understanding of different computer network architectures and standards

UNIT STANDARD NUMBER	:	114074
LEVEL ON THE NQF	:	5
CREDITS	:	5
FIELD	:	Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:	• This unit standard is intended : • To provide fundamental knowledge of the areas covered • For those entering the workplace in the area of Data Communications & Networking. • As additional knowledge for those wanting to understand the areas covered • People credited with this unit standard are able to: • Describe network types and standards • Explain local and wide area network architectures • Explain wide area network (WAN) architectures • The performance of all elements is to a standard that allows for further learning in this area.
LEARNING ASSUMED TO BE IN PLACE:	
The credit value of this unit is based on a person having prior knowledge and skills to: Demonstrate an understanding of fundamental mathematics (at least NQF level 3). Demonstrate PC competency skills (End-User Computing unit Standards, at least up to NQF level 3.). Describe data communications.	

SPECIFIC OUTCOME 1.

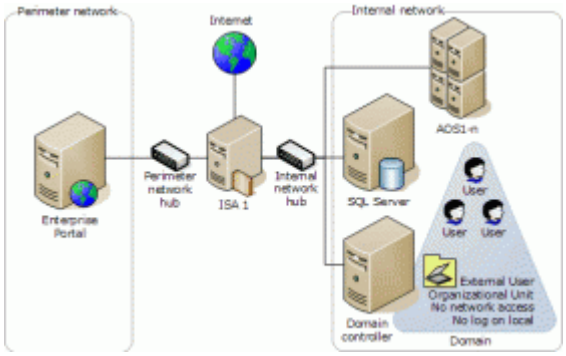
Describe computer network types and standards.

Learning Outcomes

- The description distinguishes types of networks.
- The description compares network topologies.
- The description outlines and distinguishes features of node addressing methods.
- The description outlines the standards for industry network architectures.

1.1 Describe computer network types and standards

What is network standard? Discuss different types of network standards. Which organizations define these standards?



The standards are the documents that contain technical and physical specifications about the network being designed. The networks can be reliable and efficient by following certain standards.

Types of network standards

The two types of network standards are as follows:

1. De facto standard
2. De jure standard

1. **de facto**

De facto means by tradition or by facts. These standards are developed without any formal planning. These standards come into existence due to historical developments. These standards are still being used by many organizations in the world. SNA is an example of de facto standard

2. **De Jure**

De jure means according to law or regulation. These standards are developed with proper research to fulfill the requirement of data communication. The major organization to develop communication protocols and standards are as follows:

1. American national standard institute (ANSI)
2. Institute of electrical and electronics engineers (IEEE)
3. International standard organization (ISO)
4. International telecommunications union – telecommunication standards sector (itu-t)
5. The electronic industries association (EIA)
6. Telcordia

What is network topology? Write the names of topologies.

A network can be configured or arranged in different ways. The physical layout or arrangement of connected devices in a network is called topology. It is the shape of a network. Different network topologies are as follows:

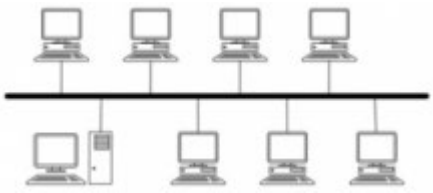
1. Bus topology
2. Ring topology
3. Star topology
4. Tree topology
5. Mesh topology

What is bus topology? Explain its working with diagram. Discuss its advantages and disadvantages.



Bus topology is the simplest type of network. It supports a small number of computers in bus topology, all computers or network nodes are connected to a common communication medium. This medium is often a central wire known as bus. The terminators are used at the end of a bus to absorb signals. A collision can occur in bus topology if two computers transmit data at same time. Bus topology is mostly used in peer-to peer networks.

Working of bus network



The sending computer sends the data and destination address through the bus. The data and address move from one computer to the other in the network. Each computer checks the address. If it matches with the address of a computer. The computer keeps the data. Otherwise the data moves to the next computer.

Advantages

1. It is simple and easy to use.
2. It requires small length of cable to connect computers.
3. It is less expensive.
4. It is easy to extend a bus. It allows more computers to join network.
5. If one node fails, it does not affect the rest of the network.

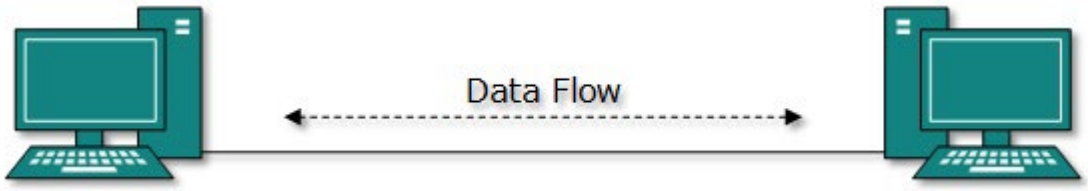
Disadvantages

1. It is difficult to troubleshoot.
2. It only supports small number of computers.
3. The network speed slows down as the number of computers increases.

A Network Topology is the way computer systems or network equipment connected to each other. Topologies may define both physical and logical aspect of the network. Both logical and physical topologies could be same or different in a same network.

Point-to-point

Point-to-point networks contains exactly two hosts (computer or switches or routers or servers) connected back to back using a single piece of cable. Often, the receiving end of one host is connected to sending end of the other end and vice-versa.

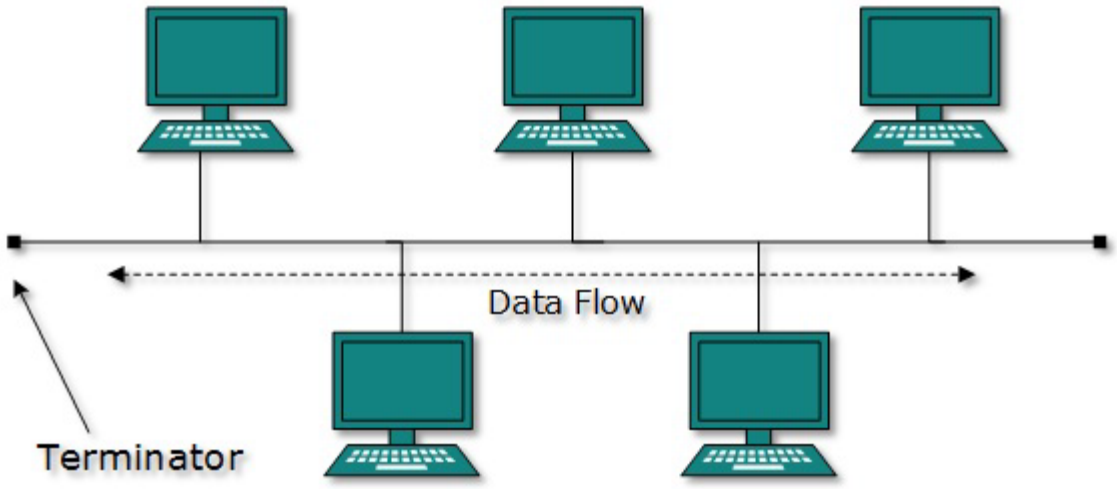


[*Image: Point-to-point Topology*]

If the hosts are connected point-to-point logically, then may have multiple intermediate devices. But the end hosts are unaware of underlying network and see each other as if they are connected directly.

Bus Topology

In contrast to point-to-point, in bus topology all device share single communication line or cable. All devices are connected to this shared line. Bus topology may have problem while more than one hosts sending data at the same time. Therefore, the bus topology either uses CSMA/CD technology or recognizes one host has Bus Master to solve the issue. It is one of the simple forms of networking where a failure of a device does not affect the others. But failure of the shared communication line make all other devices fail.

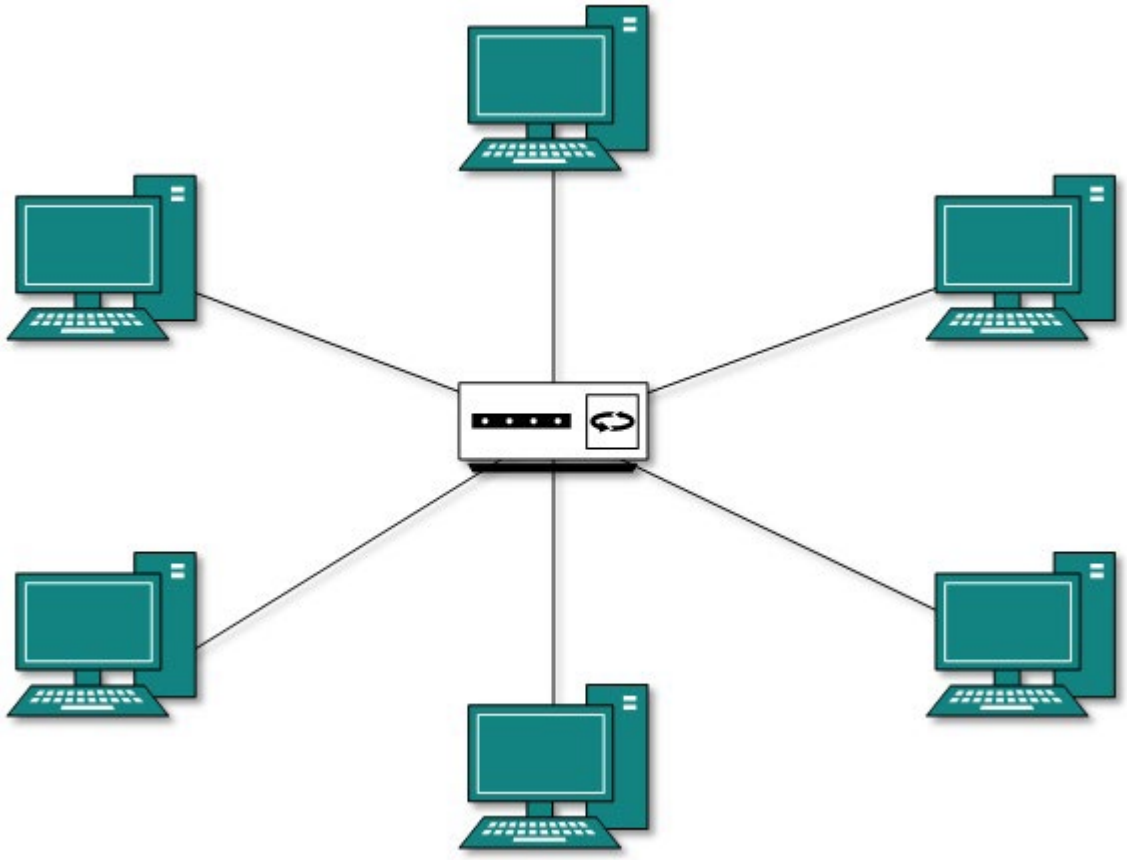


[*Image: Bus Topology*]

Both ends of the shared channel have line terminator. The data is sent in only one direction and as soon as it reaches the extreme end, the terminator removes the data from the line.

Star Topology

All hosts in star topology are connected to a central device, known as Hub device, using a point-to-point connection. That is, there exists a point to point connection between hosts and Hub. The hub device can be Layer-1 device (Hub / repeater) or Layer-2 device (Switch / Bridge) or Layer-3 device (Router / Gateway).

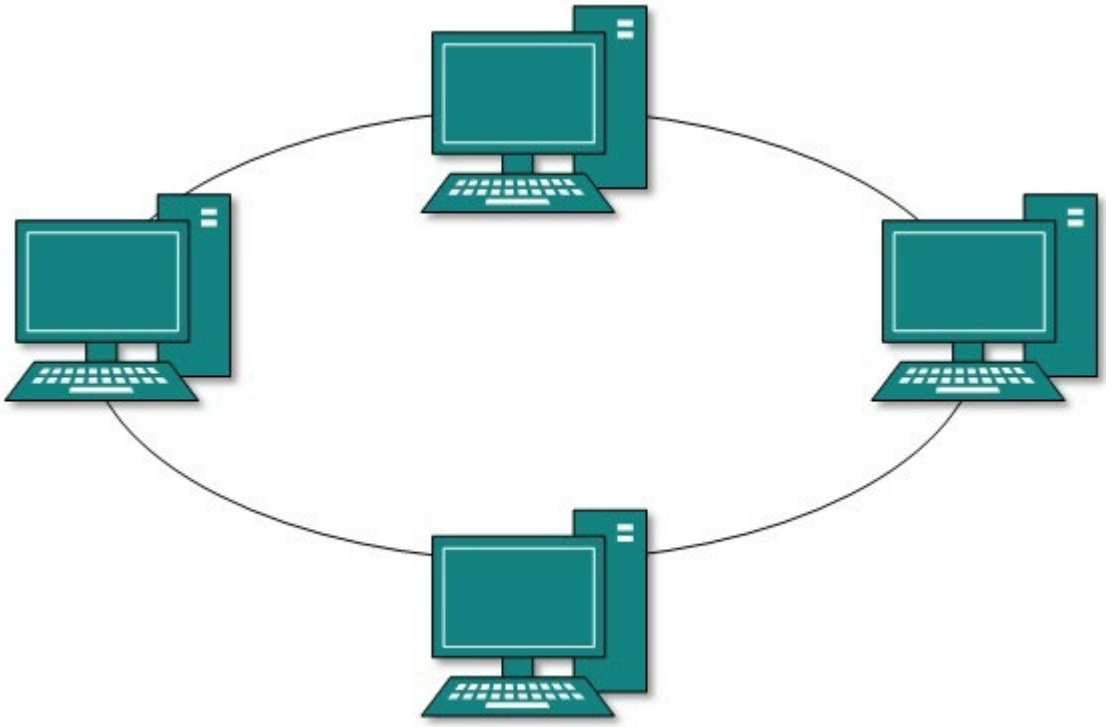


[*Image: Star Topology*]

As in bus topology, hub acts as single point of failure. If hub fails, connectivity of all hosts to all other hosts fails. Every communication happens between hosts, goes through Hub only. Star topology is not expensive as to connect one more host, only one cable is required and configuration is simple.

Ring Topology

In ring topology, each host machine connects to exactly two other machines, creating a circular network structure. When one host tries to communicate or send message to a host which is not adjacent to it, the data travels through all intermediate hosts. To connect one more host in the existing structure administrator may need only one more extra cable.

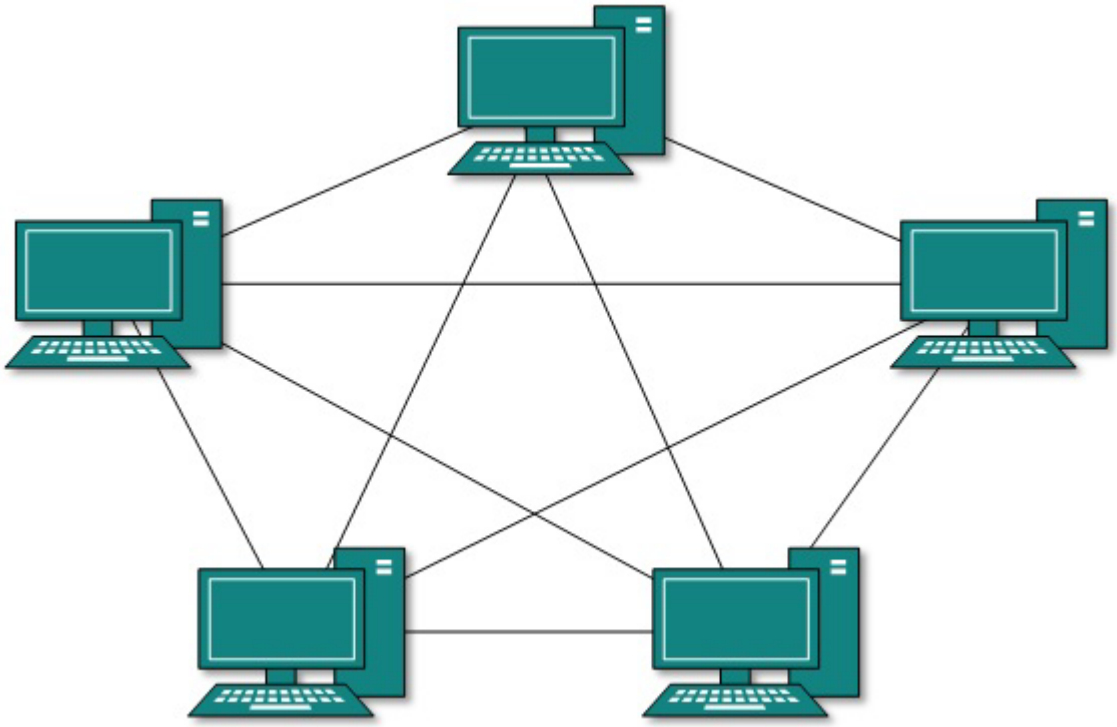


[*Image: Ring Topology*]

Failure of any host results in failure of the whole ring. Thus every connection in the ring is point of failure. There exists methods which employs one more backup ring.

Mesh Topology

In this type of topology, a host is connected to one or two or more than two hosts. This topology may have hosts having point-to-point connection to every other hosts or may also have hosts which are having point to point connection to few hosts only.



[*Image: Full Mesh Topology*]

Hosts in Mesh topology also work as relay for other hosts which do not have direct point-to-point links. Mesh technology comes into two flavors:

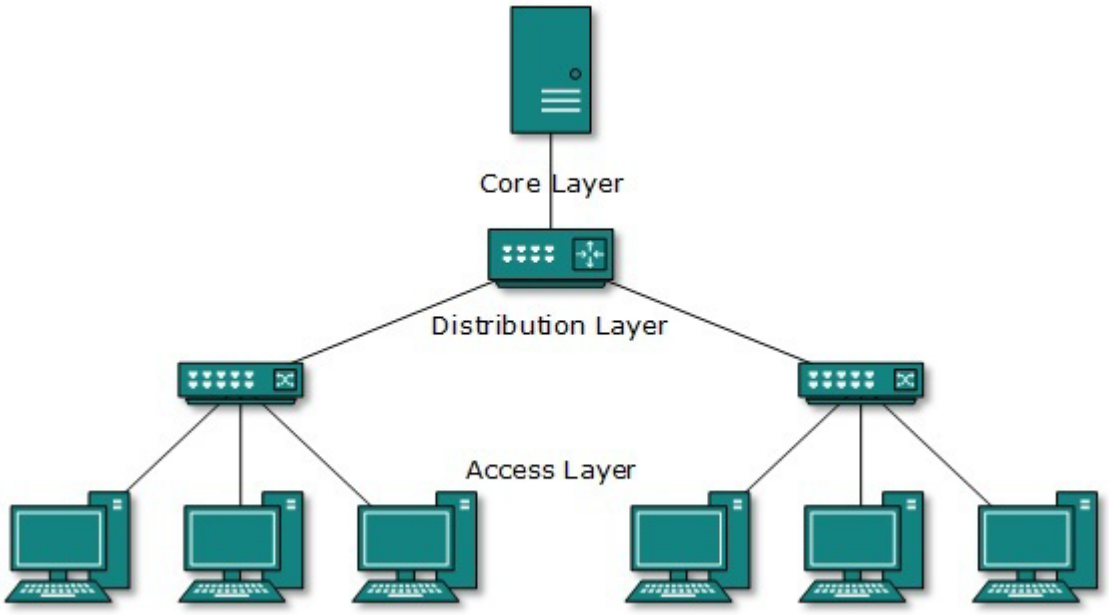
- **Full Mesh**: All hosts have a point-to-point connection to every other host in the network. Thus for every new host n(n-1)/2 cables (connection) are required. It provides the most reliable network structure among all network topologies.
- **Partially Mesh**: Not all hosts have point-to-point connection to every other host. Hosts connect to each other in some arbitrarily fashion. This topology exists where we need to provide reliability to some host whereas others are not as such necessary.

Tree Topology

Also known as Hierarchical Topology is the most common form of network topology in use present day. This topology imitates as extended Star Topology and inherits properties of Bus topology.

This topology divides the network in to multiple levels/layers of network. Mainly in LANs, a network is bifurcated into three types of network devices. The lowest most is access-layer where

user's computer are attached. The middle layer is known as distribution layer, which works as mediator between upper layer and lower layer. The highest most layer is known as Core layer, and is central point of the network, i.e. root of the tree from which all nodes fork.



[*Image: Tree Topology*]

All neighboring hosts have point-to-point connection between them. Like bus topology, if the root goes down, the entire network suffers. Though it is not the single point of failure. Every connection serves as point of failure, failing of which divides the network into unreachable segment and so on.

Daisy Chain

This topology connects all its hosts in a linear fashion. Similar to Ring topology, all hosts in this topology are connected to two hosts only, except the end hosts. That is if the end hosts in Daisy Chain are connected then it represents Ring topology.

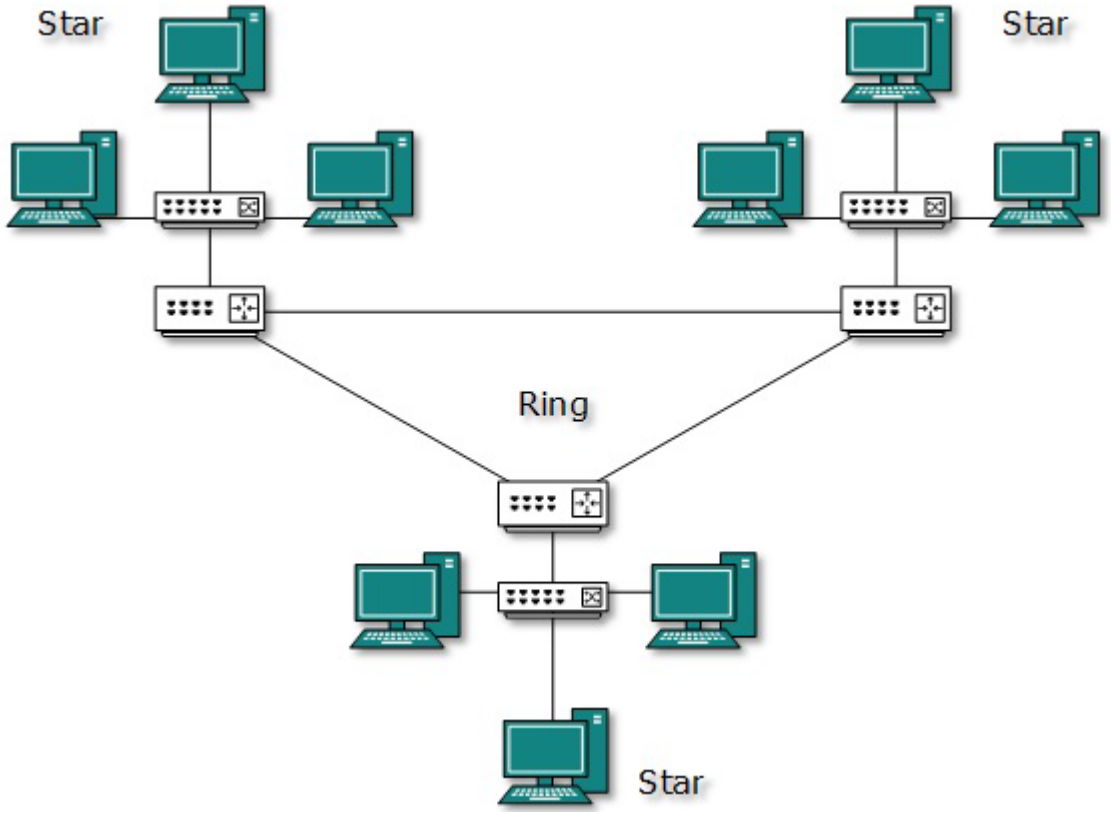


[*Image: Daisy Chain Topology*]

Each link in Daisy chain topology represents single point of failure. Every link failure splits the network into two segment. Every intermediate host works as relay for its immediate hosts.

Hybrid Topology

A network structure whose design contains more than one topology is said to be Hybrid Topology. Hybrid topology inherits merits and demerits of all the incorporating topologies.



[*Image: Hybrid Topology*]

The above picture represents an arbitrarily Hybrid topology. The combining topologies may contain attributes of Star, Ring, Bus and Daisy-chain topologies. Most WANs are connected by means of dual Ring topology and networks connected to them are mostly Star topology networks. Internet is the best example of largest Hybrid topology

SPECIFIC OUTCOME2.

The explanation outlines channel utilisation techniques and their features.

Learning Outcomes

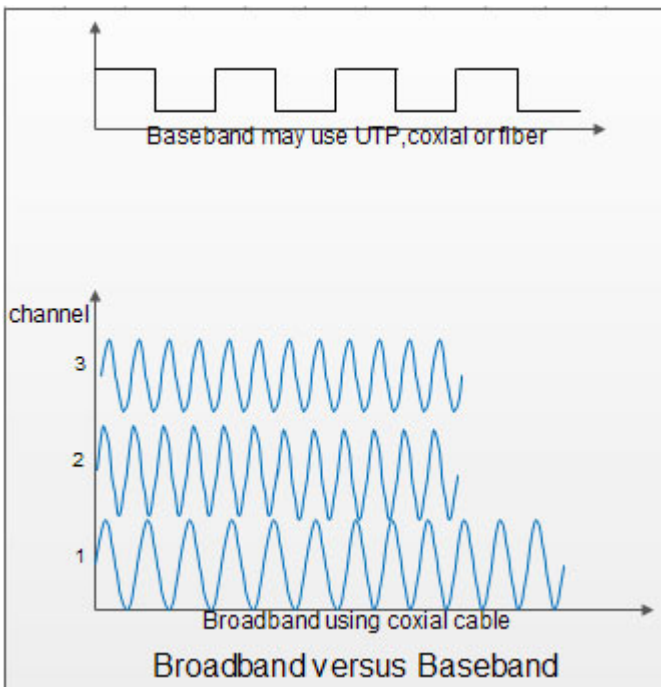
- The explanation outlines channel utilisation techniques and their features.
- The explanation compares the principles of access protocols.
- The explanation identifies the purpose of LAN components.
- The explanation illustrates network configurations and installation issues.

2.1 The explanation outlines channel utilisation techniques and their features.

Broadband A communications medium that can carry a wide range of signal frequencies, typically from audio up to video frequencies. In telecommunications the significance of a broadband system is that it can carry television and videoconferencing data as well as voice calls. A broadband medium can be made to carry many signals at once by apportioning its total bandwidth into many independent channels, each of which carries only a specific range of frequencies. In contrast, a BASEBAND can carry only a single channel. ATM, ADSL and Cable TV are all broadband media, while standard ISDN barely qualifies.

There exist two LAN transmission options, Baseband and Broadband. Baseband LANs, which is the most prevalent by far, is a Single-channel system that supports a single transmission at any given time. Broadband LANs, which are most unusual, support multiple transmissions via multiple frequency channels.

Broadband LANs



Broadband versus Baseband

Broadband LANs are Multichannel, analog LANs as shown in Figure. typically based on coaxial cable as the transmission medium, although fiber optic cable is also used. Individual channels offer bandwidth of 1 to 5 Mbps, with 20 to 30 channels typically supported. Aggregate bandwidth is as much as 500 MHz. Its characteristics are:

• Digital signal modulated onto RF carrier (analog)

• Channel allocation based on FD M

• Head-End for bidirectional transmission

• Stations connected via RF modems, Le. radio modems accomplish the digital-to-analog conversion process, providing the transmitting device access to an analog channel.

Advantages

Data, voice and video can be accommodated on broadband channel

• Greater distances

• Greater bandwidth.

Disadvantages

- Cable design
- Alignment and maintenance
- High cost, requires modems
- Lack of well-developed standards.

Some broadband LANs are referred to as IOBroadband36 where 10stands for 10Mbps, Broadband for multichannel and 36 for 3600 meters maximum separation between devices.

Baseband LANs

Baseband LAN is single channel, supporting a single communication at a time as shown in Figure. They are digital in nature. Total bandwidth of I to 100Mbps is provided over coaxial cable, UTP, STP, or fiber optic cable. Distance limitations depend on the medium employed and the specifics of the LAN protocol. Baseband LAN physical topologies including ring, bus, tree, and star.

Baseband LANs are by far the most popular and the most highly standardized. Ethernet, Token Passing, Token Ring and FDDI LANs are all baseband. They are intended only for data, as data communications is, after all, the primary reason for the existence of LANs. The characteristics of this system may be summarized as follows:

- Unmodulated digital signal
- Single channel
- Bidirectional propagation of signal
- Stations connected, via T connectors
- No need of modems - low cost installation.

Advantages

- Simplicity
- Low cost
- Ease of installation and maintenance
- High rates.

Disadvantages

- Limited distances
- Data and voice only.

6 Required Components of a LAN

A Local Area Networks connects computers together to exchange data. Apart from the computers, and other devices like printers and faxes, a LAN has to have six essential components to function.

Network Adapter

A computer needs a network adapter to connect to a network. It converts computer data into electronic signals. It listens for silence on the network cable and applies the data to it when it has an opportunity. The network access element of its job is called Media Access Control, or MAC. The physical address of every computer on a network is called its MAC address. The MAC address is the network adapter's serial number. Most computers are shipped with the network adapter integrated into the motherboard. However, early PCs didn't include this function and computer owners had to buy it separately and fit it into an expansion slot on the motherboard. These were called "network cards" because they were sold on a separate card. Although network adapters are now integrated, the name network card is still used. The wireless equivalent is called a Wireless Network Interface Controller.

Network Medium

Wired networks need cable. The most common form of cable used in networks is called the "Unshielded Twisted Pair." In PC shops, it is generally just referred to as "network cable" or "Ethernet cable." Ethernet is the most widely implemented set of standards for the physical properties of networks. UTP is so closely identified with Ethernet that it is often given that name. Other cable types used for networks are twin-axial, Shielded Twisted Pair and single-mode and multi-mode fiber optic cable. Wireless networks don't need cable; they send data on radio waves generated by the WNIC.

Cable Connectors

In wired networks, the most common form of connector is the RJ45. Every computer with networking capabilities has an RJ45 port. This is sometimes called a "network port" or an "Ethernet port." The RJ45 plug looks like a slightly larger telephone plug and connects the Unshielded Twisted Pair or the Shielded Twisted Pair cable.

Power Supply

Both wired and wireless networks need a power supply. A wireless network uses the current to generate radio waves. A cabled network sends data interpreted as an electronic pulse.

Hub/Switch/Router

In wired networks, one computer cannot connect to many others without some form of splitter. A hub is little more than a splitter. It repeats any signals coming into one of its ports out onto all its other ports. A cable leads from each port to one computer. A switch is a more sophisticated version of a hub. It only sends the signal on to the computer with the address written in the arriving message. Routers are much more complicated and are able to forward messages all over the world. Larger networks sometimes use routers for their LAN traffic. The wireless networking device is called a "wireless router."

Network Software

Software on a communicating computer packages data into segments and puts that data into a structure called a "packet." The source and destination addresses of the packet are written into the header of the packet. The receiving computer needs to interpret these packets back into meaningful data and deliver it to the appropriate application.

Ethernet

Ethernet is a Local Area Network implementation technology which is widely deployed. This technology was invented by Bob Metcalfe and D.R. Boggs in early 70s. It was standardized in IEEE 802.3 in 1980. Ethernet is network technology which shares media. Network which uses shared media has high probability of data collision. Ethernet uses CSMA/CD technology to detect collisions. CSMA/CD stands for Carrier Sense Multi Access/Collision Detection. When a collision happens in Ethernet, all its host rolls back and waits for some random amount of time and then re-transmit data.

Ethernet connector, i.e. Network Interface cards are equipped with 48-bits MAC address. This help other Ethernet devices to identify and communicate with remote devices in Ethernet.

Traditional Ethernet uses 10BASE-T specifications. 10 is for 10mpbs speed, BASE stands for using baseband and T stands for Thick net or Thick Ethernet. 10BASE-T Ethernet provides transmission speed up to 10mbps and uses Coaxial cable or Cat-5 Twisted Pair cable with RJ-5 connector. Ethernet follows Star Topology with segment length up to 100 meters. All devices are connected to a Hub/Switch in a Star Fashion.

Fast-Ethernet

To encompass need of fast emerging software and hardware technologies, Ethernet extends itself as Fast-Ethernet. It can run on UTP, Optical Fiber and can be wireless too. It can provide speed up to 100 mbps. This standard is named as 100BASE-T in IEEE 803.2 using Cat-5 Twisted pair cable. It uses CSMA/CD technique for wired media sharing among Ethernet hosts and CSMA/CA (Collision Avoidance) technique for wireless Ethernet LAN.

Fast Ethernet on fiber is defined under 100BASE-FX standard which provides speed up to 100mbps on fiber. Ethernet over Fiber can be extended up to 100 meters in half-duplex mode and can reach maximum of 2000 meters in full-duplex over multimode fibers.

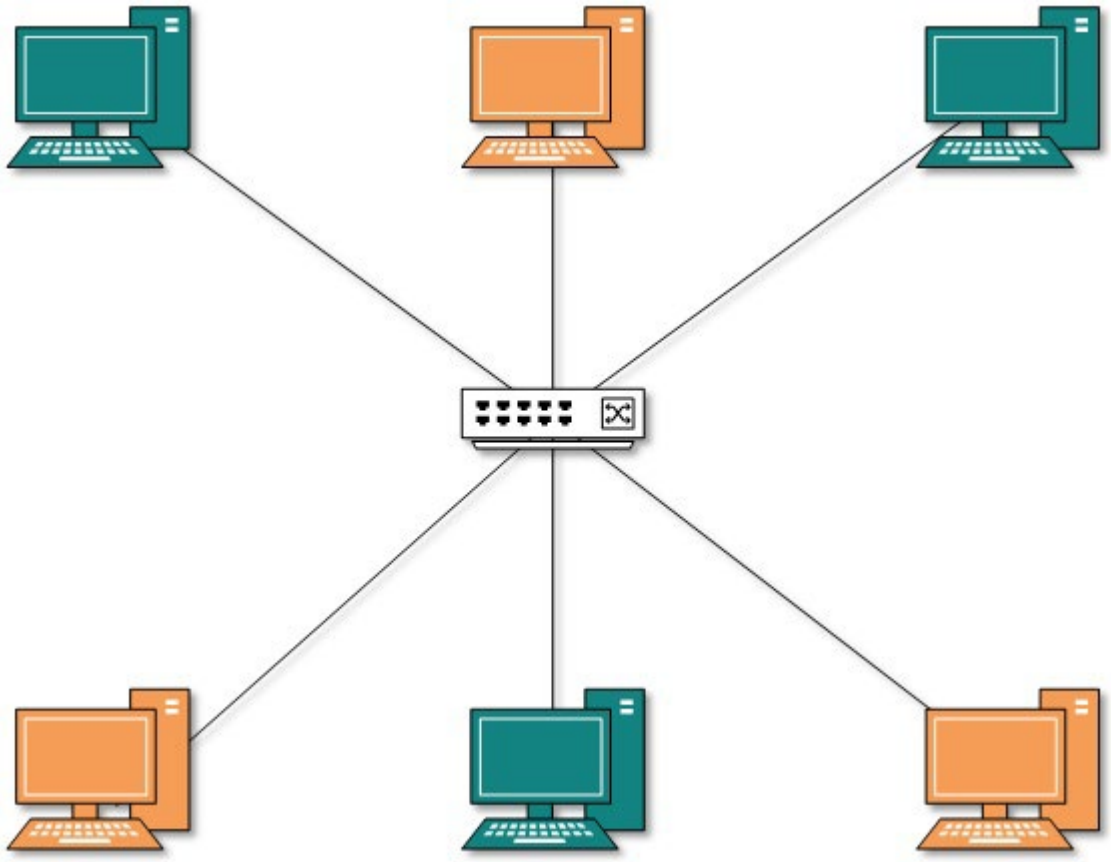
Giga-Ethernet

After being introduced in 1995, Fast-Ethernet could enjoy its high speed status only for 3 years till Giga-Ethernet introduced. Giga-Ethernet provides speed up to 1000 mbits/seconds. IEEE802.3ab standardize Giga-Ethernet over UTP using Cat-5, Cat-5e and Cat-6 cables. IEEE802.3ah defines Giga-Ethernet over Fiber.

Virtual LAN

LAN uses Ethernet which in turn works on shared media. Shared media in Ethernet create one single Broadcast domain and one single Collision domain. Introduction of switches to Ethernet has removed single collision domain issue and each device connected to switch works in its separate collision domain. But even Switches cannot divide a network into separate Broadcast domain.

Virtual LAN is a method to divide a single Broadcast domain into more than one Broadcast domains. Host in one VLAN cannot speak to a host in another. By default, all hosts are placed into same VLAN.



[*Image: Virtual LAN*]

In above pictures, different VLANs are depicted in different color codes. Hosts in one VLAN, even if connected on the same Switch cannot see or speak to other hosts in different VLANs. VLAN is Layer-2 technology which works closely on Ethernet. To route packets between two different VLANs a Layer-3 device (such as Router) is required.

SPECIFIC OUTCOME 3.

Explain LAN architectures.

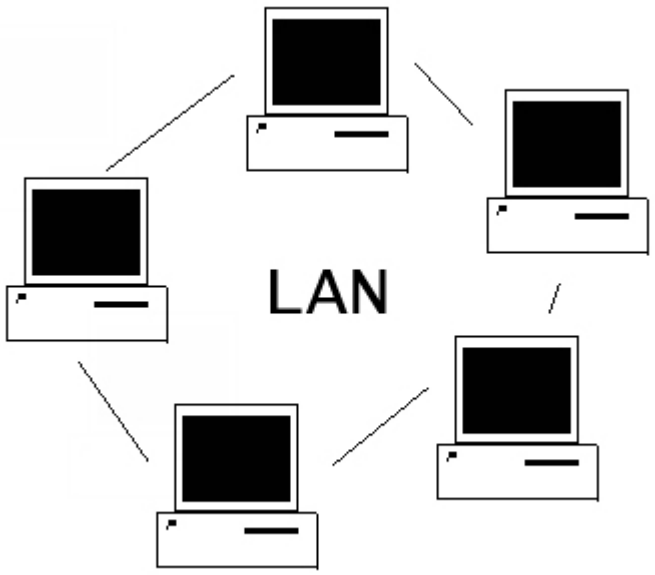
Learning Outcomes

- 1. The explanation identifies switching techniques.
- 2. The explanation compares switching techniques.
- 3. The explanation outlines WAN management issues.

3.1 Explain LAN architectures.

What is network architecture?

Network architecture refers to the layout of the network, consisting of the hardware, software, connectivity, communication protocols and mode of transmission, such as wired or wireless. Know about the types of network classified according to the areas covered such as LAN, MAN and WAN. Learn about the network topologies categorized according to the layout of equipments and computers such as star, loop, bus, or mesh topologies. There are many communication protocols used in the networking technology. It is important to know about the network architecture as networks play a very important role in today's world



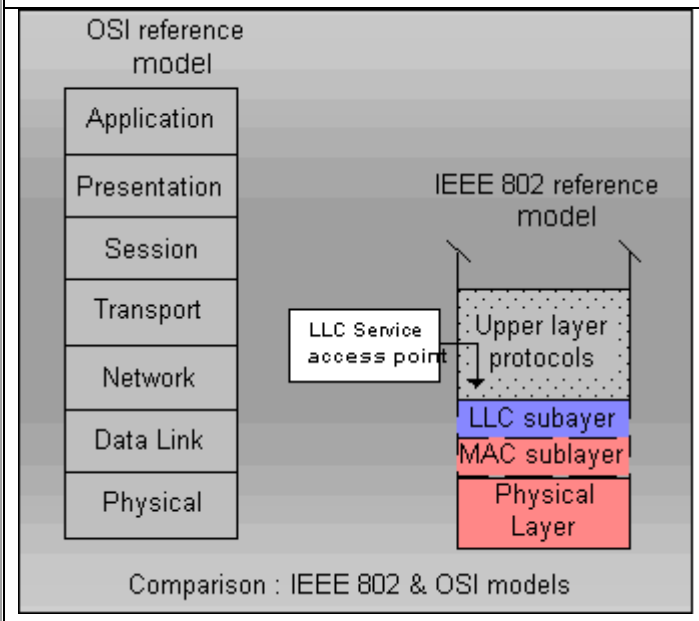
LAN ARCHITECTURE

Protocols needed to transfer data frames among stations connected on the same physical link in a local area are data link control protocols. The original specifications of the OSI's Data Link Control protocol standard, the HDLC, predates the advent of local area networks (LANs). HDLC has been designed for connecting remotely located DTEs connected with links that are relatively lower data rates and more likely to subject to distortion and noise, while the protocols for LANs need to access much higher data rates, possibly over a more reliable physical medium. To achieve this objective, the protocol standard organization divide the Data Link Control layer into two sublayers, with the physical medium access and the data transfer functions being specified as a new sublayer, the MAC (Media Access Control), that can meet the need of a LAN environment. MAC replaces the physical medium access and the data transfer functions that would otherwise be performed by HDLC. The remaining functions of HDLC are performed by another new sublayer, the LLC (Logical Link Control).

Under IEEE LAN standards, all LAN protocols share the common LLC specifications, but vary with MAC and physical layer specifications. The standard for LLC is designated as IEEE 802.2. One service being performed by the LLC is the access to the network layer. LLC is able to

perform other services that are similar to HDLC's services. But in actual implementations, these other services are rarely employed.

Among the popular IEEE LAN standard specifications are 802.3 for the Ethernet, 802.5 for the token ring and 802.11 for the wireless LAN.



Comparison : IEEE 802 & OSI models

LAN architecture

The LAN architecture consists of three layers: Physical, MAC (Medium Access Control) and LLC (Logical Link Control).

- LLC provides connection management, if needed. (For most applications, it is not needed.)
- MAC is a protocol for accessing high speed physical links and for transferring data frames from one station to another.
- Physical layer deals mainly with actual transmission and reception of bits over the transmission medium. Its specification depends on the specific physical medium and MAC protocols it interfaces with.

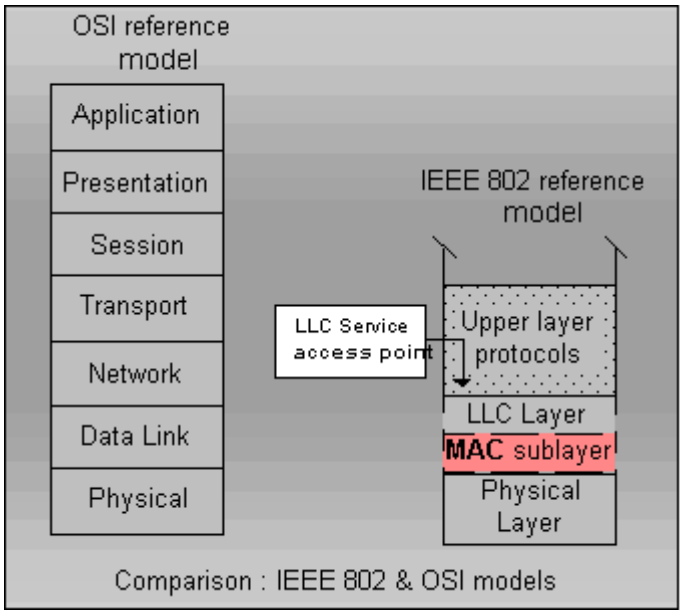
In reference to the OSI Reference Model, the Physical layer is the same. The LLC and MAC are sublayers of the DLC layer.

LAN ARCHITECTURE -->**MAC functions**

A MAC entity performs the following three functions:

- controlling access to the transmission medium,
- source and destination station addressing and
- frame error detection.

A typical MAC entity provides an unacknowledged connectionless, datagram service. For most applications, LAN protocols provide services to an IP entity. Since the IP only needs an unacknowledged datagram service, no connection management service is needed from LLC.



LAN ARCHITECTURE -->**MAC Frame Format**

Shown below is a MAC frame that contains MAC's key fields. Actual MAC frame format vari different LAN standards.

Variable	6 octets	6 octets	Variable	4 octets

MAC Control	DA	SA	LLC PDU	CRC

The meaning and functions of the frame fields are explained in the chart below.

MAC Control	MAC Control	Control information for MAC protocol. Variable in length and contents. Depending on specific LAN Standards.
DA	Destination Address	Destination physical attachment point on the LAN's transmission media.
SA	Source Address	Source physical attachment point on the LAN's transmission media. Each MAC address is unique and burned in at the factory.
LLC PDU	LLC Protocol Data Unit	Data from LLC sublayer.
CRC	Cyclic Redundancy Check	Same as FCS (Frame Check Sequence) field in HDLC. This is used for error checking. If an error is detected, the frame is simply discarded. MAC protocol does not recover erred frames.

LAN ARCHITECTURE -->**LLC Frame Format**

Shown below is an LLC PDU that contains LLC's fields.

	1 octet	1 octet	1 or 2 octets	variable size
LLC PDU	**DSAP**	**SSAP**	**LLC Control**	**Information**

and functions of the frame fields are explained in the chart below.

DSAP	Destination Service Access Point	DSAP is the address of the network layer entity that uses the LLC service. It identifies the network layer protocol the packet is destined for. It is also known as the port number. Multiple DSAPs may be multiplexed on a single MAC address.
SSAP	Source Service Access Point	SSAP is the address of the network layer entity that uses the LLC service. It identifies the network layer protocol the packet comes from. It is also known as the port number. Multiple SSAPs may be multiplexed on a single MAC address.
LLC Control	LLC Control	Similar to HDLC's Control field, but with field parameters set to the values that are applicable to the LAN environment.(See last section, "LLC Service").
Information	Information	PDU from LLC's users, such as IP packets.